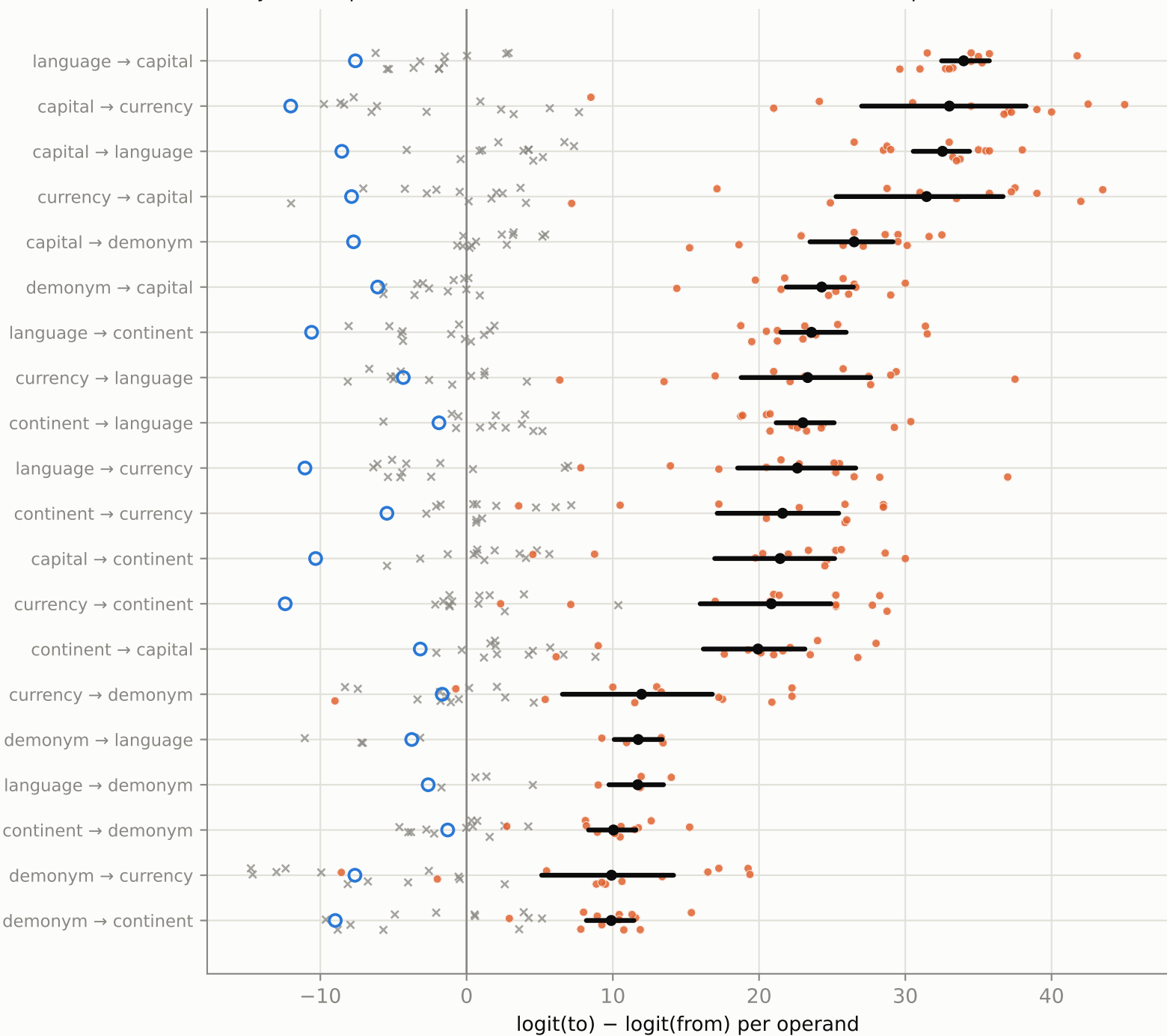


Every ordered operator swap flips, with uncertainty shown

family level (operators as clusters): contrast +22.6 [+14.0, +32.1] · flip fraction 100% [100%, 100%]



dots = swap per operand (orange, 12/pair) · gray x = matched-norm random · bar = operand-bootstrap 95% CI · ○ = clean baseline mean
 20 ordered pairs share 5 operator directions — pairs are not independent; operands are the unit within a pair, operators across the paradigm.